Inspection fixture - design brief

AI BRAKING / AB-MON / C0 / 2026-09-07

A non-safety bench fixture for checking a disc with bought supports and a movable indicator arm. It gives inspection a repeatable physical datum while electronics and AI stay at requirements level.

TARGETS (not validated)

Supported disc: 400 mm concept. Packaging check

Runout study target: ≤0.05 mm. Proposed requirement, not a part acceptance limit

Gauge resolution objective: ≤0.005 mm. Ten-to-one study rule; full uncertainty needed

Acceptance authority: Calibrated measurement + approved plan. AI advisory only

DESIGN DECISIONS

Buy calibrated indicators and supports; fabricate the base and adjustable brackets.

Separate model anomaly detection from final dimensional acceptance.

Tie each reading to part, fixture, instrument, calibration and environmental records.

CANDIDATE MATERIALS

Base: machined steel candidate; stiffness and stability require assessment.

Arms: aluminium 6082-T6 candidate; non-load-bearing metrology supports.

Indicators and centres: purchased calibrated equipment.

Assembly and engineering gates

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ASSEMBLY INTENT

1. Level and qualify the base and purchased centres.
2. Mount the rotor without distorting its datum surfaces.
3. Set a calibrated probe normal to the inspected face.
4. Record repeated rotations, remounts and operator variation before acceptance use.

OPEN DESIGN RISKS

1. No measurement-system analysis or uncertainty budget has been completed.
2. Fixture and datum errors can dominate small runout readings.

VALIDATION REQUIRED

1. Gauge repeatability/reproducibility and uncertainty budget.
2. Comparison with calibrated reference artefacts and independent CMM measurement.

ELECTRICAL / SENSOR REQUIREMENTS ONLY

1. Define displacement range/resolution, triggering, sampling, DAQ, traceable calibration, environmental sensing, storage and human approval workflow.

Assembly reference / numbered components

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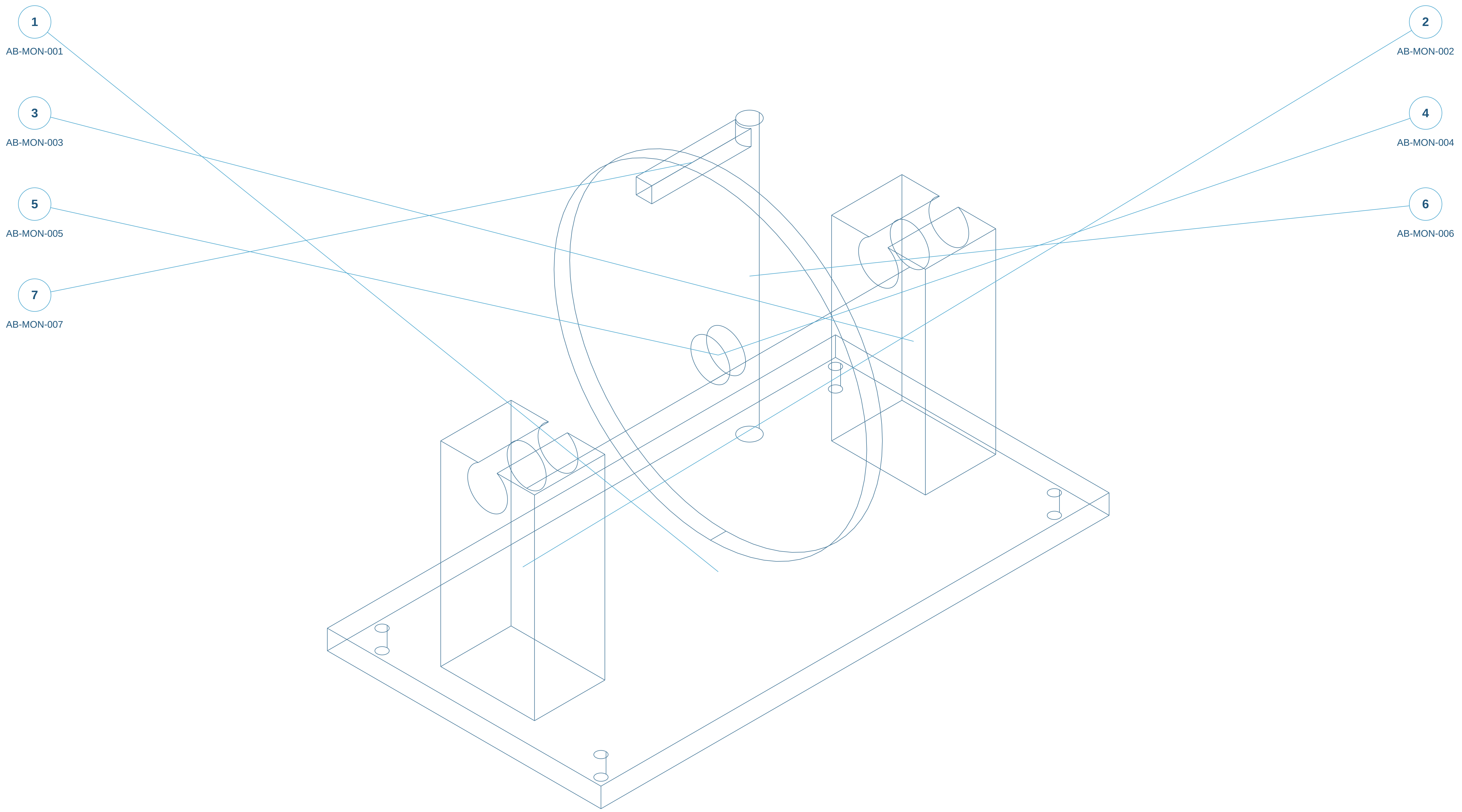

Static isometric reference with item marks. Exploded geometry and part selection are available in the web workspace.

ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

Parts and interfaces

AB-MON-001 | Inspection base | make-concept

S355J2+N candidate; condition and design properties unapproved; Saw/laser blank; CNC critical faces and bores

AB-MON-002 | Centre support envelope -1 | bought-envelope

Supplier-controlled; illustrative reserved envelope; Buy precision centre/support

AB-MON-003 | Centre support envelope 1 | bought-envelope

Supplier-controlled; illustrative reserved envelope; Buy precision centre/support

AB-MON-004 | Study mandrel | make-concept

42CrMo4 candidate; heat treatment / fit unapproved; Turn/grind after datum and mounting design

AB-MON-005 | Reference disc | interface-envelope

Reference test piece; not AB-ROT interface; Reference only

AB-MON-006 | Indicator column | make-concept

6082-T6 aluminium candidate; Machine adjustable metrology support

AB-MON-007 | Probe arm envelope | bought-envelope

Supplier-controlled; illustrative reserved envelope; Buy articulated holder and calibrated indicator

AB-MON-001 / Inspection base

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TOP X-Y

S355J2+N candidate; condition and design properties unapproved

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 650.00 / 350.00 / 25.00 mm

650 x 350 x 25; 4 x diameter 13 at (+/-290,+/-140).

Flatness datum and levelling scheme not released.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

FRONT X-Z

SIDE Y-Z

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AB-MON-002 / Centre support envelope -1

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TOP X-Y

Supplier-controlled; illustrative reserved envelope

Buy precision centre/support

Overall X/Y/Z bounds: 90.00 / 120.00 / 250.00 mm

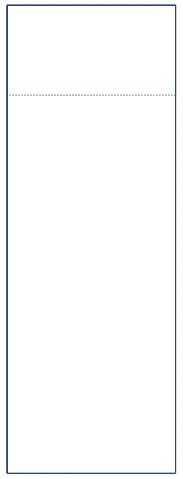
Reserved supports; spindle axis Z 240.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

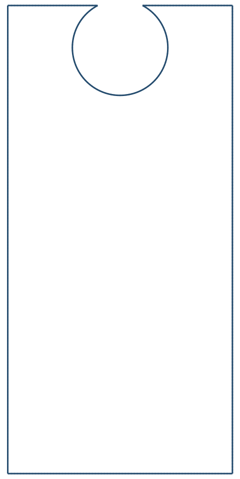
Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

FRONT X-Z



SIDE Y-Z



AB-MON-003 / Centre support envelope 1

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Supplier-controlled; illustrative reserved envelope

Buy precision centre/support

Overall X/Y/Z bounds: 90.00 / 120.00 / 250.00 mm

Reserved supports; spindle axis Z 240.

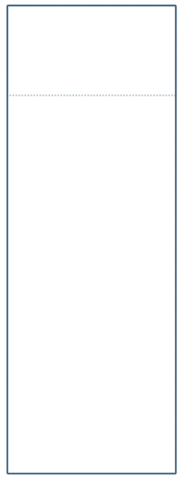
CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

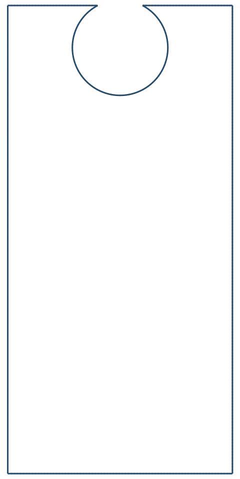
Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

TOP X-Y

FRONT X-Z



SIDE Y-Z



AB-MON-004 / Study mandrel

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TOP X-Y

42CrMo4 candidate; heat treatment / fit unapproved

Turn/grind after datum and mounting design

Overall X/Y/Z bounds: 490.00 / 50.00 / 50.00 mm

Diameter 50 nominal; no centre holes, bearing seats or expanding feature.

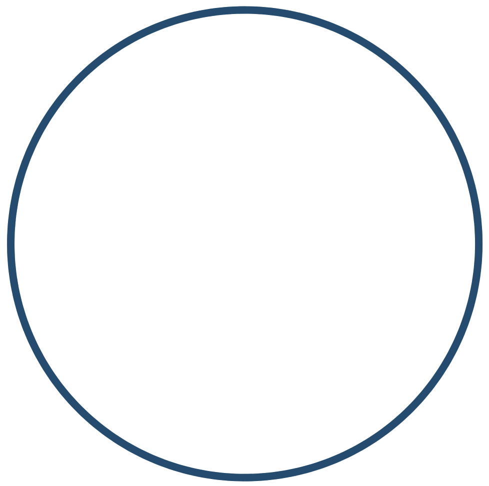
CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

FRONT X-Z

SIDE Y-Z



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AB-MON-005 / Reference disc

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Reference test piece; not AB-ROT interface

Reference only

Overall X/Y/Z bounds: 20.00 / 400.00 / 400.00 mm

All dimensions nominal. Fits, GD&T and edge details awaiting engineer release.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

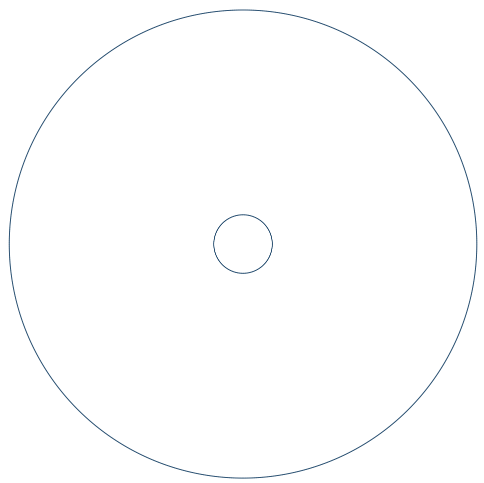
Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

TOP X-Y

FRONT X-Z

SIDE Y-Z



AB-MON-006 / Indicator column

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6082-T6 aluminium candidate

Machine adjustable metrology support

Overall X/Y/Z bounds: 25.00 / 25.00 / 350.00 mm

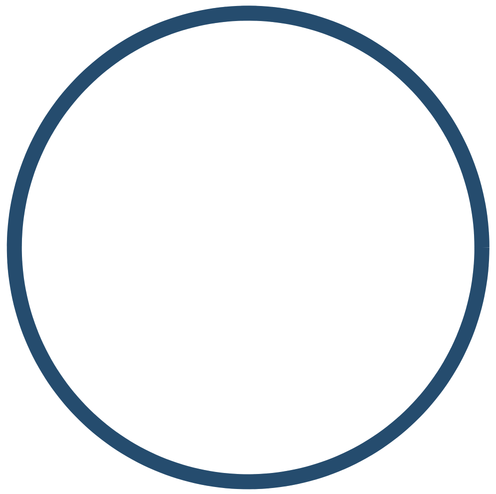
All dimensions nominal. Fits, GD&T and edge details awaiting engineer release.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

TOP X-Y



FRONT X-Z

SIDE Y-Z

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AB-MON-007 / Probe arm envelope

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TOP X-Y

Supplier-controlled; illustrative reserved envelope

Buy articulated holder and calibrated indicator

Overall X/Y/Z bounds: 127.09 / 20.00 / 20.00 mm

All dimensions nominal. Fits, GD&T and edge details awaiting engineer release.

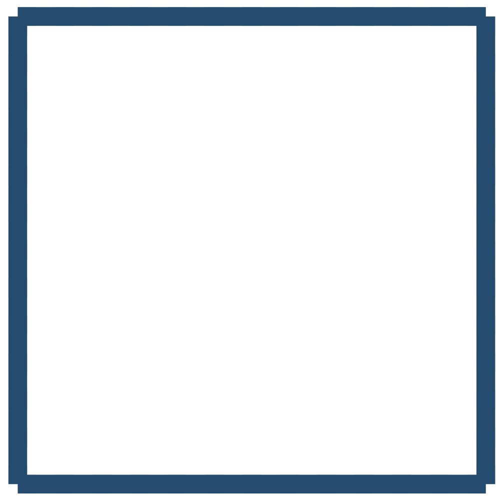
CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

FRONT X-Z

SIDE Y-Z



ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.